S1.3 Electron configurations

The big picture

Guiding question(s)

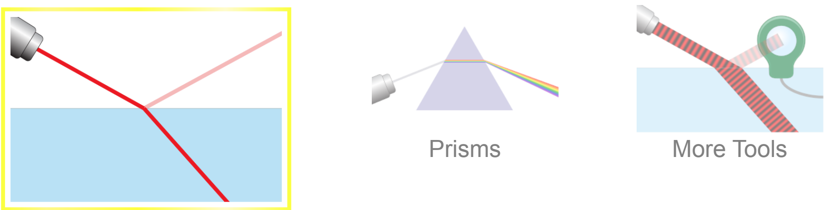
How can we model the energy states of electrons in atoms?

Keep the guiding questions in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Colours are all around us in our daily lives. From the blue colour of the sky to the green colour of the leaves on the trees. The reason we see different colours is related to a particular region of the electromagnetic spectrum, specifically the visible region. All of the natural light that reaches the Earth comes from the Sun, which is also the source of most of our heat. Light from the Sun is known as white light. You can observe the different colours that make up white light with this simple experiment.

1. Collect a flashlight, some masking tape, a clear glass and a piece of white paper.
2. Cover the flashlight with the masking tape leaving a thin slit for the light to shine through.
3. Fill the glass with water and place it on the white paper. Dim the lights in the room.
4. Hold the flashlight so the slit is vertical and shine the flashlight through the water at an angle until you see the different colours of light on the paper.
5. Make a note of the colours and the order in which they appear.

Alternatively, a simulation that allows you to see the colours of white light below. Click on the Prisms icon and then select the third beam icon on the right-hand side of the screen. Press the red button on the torch to get a beam of white light and then pick up the triangular prism and place it into the beam. You will see that as the beam exits the prism it has been split into the colours of the visible spectrum.



Interactive 1. Bending light simulation.

More information for interactive 1

The interactive simulation titled "Bending light simulation," explores how light bends when transitioning between different materials with varying refractive indices. A laser emits a beam of light at an interface between two media, demonstrating refraction and reflection. Users can measure the angles of incidence and transmission with a simulated protractor or by enabling the "angle" option in the bottom left. The simulation allows adjustment of the material properties, including selecting a "mystery" medium, challenging users to determine its refractive index using Snell's law. The speed of light in each medium can be observed, providing insights into how refraction relates to changes in wave velocity.

Users can also modify the wavelength of the incident light, exploring whether this affects refraction. The interface includes tools to measure intensity and speed, allowing a more quantitative approach to understanding light behavior. The first medium is typically air, while the second medium can be changed to water, glass, or a mystery substance. The refractive index of known materials is displayed, while the mystery medium requires calculation. Set the first medium as "air" and the second medium as "mystery A" by adjusting the slider in the "material" boxes.

Set the angle of incidence, $\theta_i = 60^\circ$. The angle of refraction as per the simulation will display, $\theta_r = 21^\circ$. According to Snell's law, we have:

$$n_a \sin\theta_i = n \sin\theta_r$$

The value of n from the above equation is:

$$n = \frac{n_a \sin \theta_i}{\sin \theta_r}$$

Substitute the values in the above equation to determine the refractive index of the "mystery A" medium.

$$n = \frac{1 \times \sin 60^\circ}{\sin 21^\circ}$$

$= 2.416$

Thus, the refractive index of the "mystery A" medium is 2.416. In this way, the interactive helps to determine the refractive index of a medium.

Enabling the "wave" mode and adjusting the slider on the top right, users can experience the change in wavelength due to the change in speed of the wave. The increase in the index of refraction decreases the speed of light. Since the frequency of light is a constant, the wavelength should decrease.

The light beam can be switched between ray and wave representations, reinforcing the wave nature of light and how it interacts at boundaries. The near-perfect data accuracy contrasts with real experiments, where measuring instruments introduce variability. Despite this limitation, the simulation effectively visualizes refraction, total internal reflection, and the relationship between angle, speed, and refractive index. By adjusting different parameters and analyzing results, users develop a deeper understanding of light's interaction with materials.

The colours you will see are the colours that make up white light. Interestingly, white light is in fact not one colour but a range of colours which our eyes view as white.

The sub-atomic particle in the atom responsible for the absorption or emission of light is the electron. As you will learn in this section, electrons can absorb and emit energy as they move between the energy levels of an atom. Not only does this give rise to colour but also provides evidence for the existence of discrete energy levels in the atom.

Prior learning

Before you study this subtopic make sure that you understand the following (see section S1.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/)):

- Atoms contain a dense positively charged nucleus that contains the protons and neutrons.
- Negatively charged electrons occupy the space outside the nucleus.
- How to determine the number of sub-atomic particles in an atom or ion.